Project Design Phase
Proposed Solution Template

Date	25 JULY 2026
Team ID	
Project Name	CAB BOOKING
Maximum Marks	2 Marks

Parameter	Description
Problem Statement (Problem to be solved)	Traditional cab booking methods are often time-consuming, lack fare transparency, provide no real-time ride tracking, and require manual booking. Customers face delays in finding nearby cabs, while administrators struggle to manage bookings and drivers efficiently.
Idea / Solution Description	Develop a Cab Booking System using the MERN Stack (MongoDB, Express.js, React.js, Node.js) that enables users to register, search nearby cabs, estimate fares, book rides, track drivers in real time, make secure online payments, and manage booking history. The system also provides dedicated dashboards for drivers and administrators.
Novelty / Uniqueness	The system combines role-based authentication, real-time ride tracking, fare estimation, secure JWT authentication, online payment integration, and responsive web design into a single platform, providing a seamless experience for customers, drivers, and administrators.
Social Impact / Customer Satisfaction	The application improves transportation accessibility by reducing booking time, enhancing passenger safety through live tracking, providing transparent pricing, and offering a convenient digital platform. It also creates better earning opportunities for drivers and improves service management for administrators.
Business Model	Revenue can be generated through commission on each completed ride, driver registration fees, premium memberships,

Parameter	Description
(Revenue Model)	advertisements, subscription plans for fleet owners, and service charges on online payments.
Scalability of the Solution	The MERN Stack architecture makes the system highly scalable. Future enhancements such as AI-based driver matching, ride scheduling, ride sharing (carpooling), push notifications, mobile applications, multiple payment gateways, and cloud deployment can be integrated with minimal changes to the existing system.